

The process is as follows:

- The Cuckoo central component runs in the host machine. This central component is the pivot around which the entire analysis process revolves. The optimal configuration for the Cuckoo host is Ubuntu.
- The malware files are analysed inside guests, which are protected virtual environments. The configuration for guests could be Windows version 7, Vista, etc. It supports Mac OS X as well.
- For each malware file submitted for analysis, a fresh virtual machine is launched and analysis is done in that protected environment.
- The host machine accesses the results of the analysis process in a safer manner.

Cuckoo installation

As of now, Cuckoo doesn't come as a simple click-and-install application. The system needs to be configured with many dependencies, libraries, etc. Though the installation is a bit lengthy, it is worth spending time on it, if you require its malware analysis capabilities for your organisation. The installation process can be divided into two major phases:

- Preparing the host
- Preparing the guest

Preparation of the host

The host preparation involves installing the following libraries and software for the effective functioning of Cuckoo:

- Python libraries
- Virtualisation software
- For network analysis, tcpdump is needed
- To enable memory analysis, Volatility (<https://code.google.com/p/volatility/>), an advanced memory forensic framework, is needed
- The Cuckoo sandbox is required for the core analysis component
- Various *.conf* files like *cuckoo.conf*, *memory.conf*, etc, need to be configured

As Cuckoo is developed using Python, the host machine should have Python installed on it. The recommended Python version is 2.7, which can be installed using the following command:

```
$ sudo apt-get install python
```

If the Web interface that is based on Django is required, then MongoDB needs to be installed, as follows:

```
$ sudo apt-get install mongodb
```

Cuckoo needs SQLAlchemy and BSON, which can be installed using the following command:

```
$ sudo apt-get install python-sqlalchemy python-bson
```

To analyse the network activities, Cuckoo needs tcpdump, which can be installed as follows:

```
$ sudo apt-get install tcpdump
```

The libraries listed below are not absolutely mandatory but they would make Cuckoo's analysis much more effective, if installed.

- Dpkt – Python packet creation/parsing library
 - Jinja2 for HTML reports
 - Yara for signature matching in Python
 - MAEC (malware attribute enumeration and characterisation)
 - Magic library for file format identification
 - Volatility for memory forensic analysis which can be acquired from <https://github.com/volatilityfoundation>
- For the virtualisation software, VirtualBox is preferred. Though Cuckoo supports other virtualisation tools as well, the association between VirtualBox and Cuckoo goes back to earlier versions, which makes it the preferable tool. VirtualBox can be acquired from <https://www.virtualbox.org/wiki/Downloads>. VirtualBox binaries are available for Windows, OS X, Linux and Solaris hosts.

For the seamless functioning of Cuckoo, creating a separate user is preferred in the official documentation. The latest version of Cuckoo can be downloaded from the official website (<http://www.cuckoosandbox.org/download.html>).

Cuckoo configuration is done using six different configuration files as listed below. The detailed information on each of these files' configurations is available at <http://docs.cuckoosandbox.org/en/latest/installation/host/configuration/>.

- *cuckoo.conf* for the general analysis process
- *auxiliary.conf* for handling optional modules
- *machinery.conf* for virtualisation software options
- *memory.conf* for configuring Volatility
- *processing.conf* for the settings of processing modules
- *reporting.conf* for various report formats

Preparation of the guest

As the malware analysis is performed on the virtual machines, they need to be prepared properly. After preparation, they need to be configured as explained in the following instructions:

- Python needs to be installed on the guests.
- *Python Image Library* is also preferred for making screen snaps when the analysis is going on.
- The Windows firewall and automatic updates need to be disabled, failing which the behaviour of the malware may be affected, which would reduce the power of the analysis on the network front.
- Virtual networking should be configured for